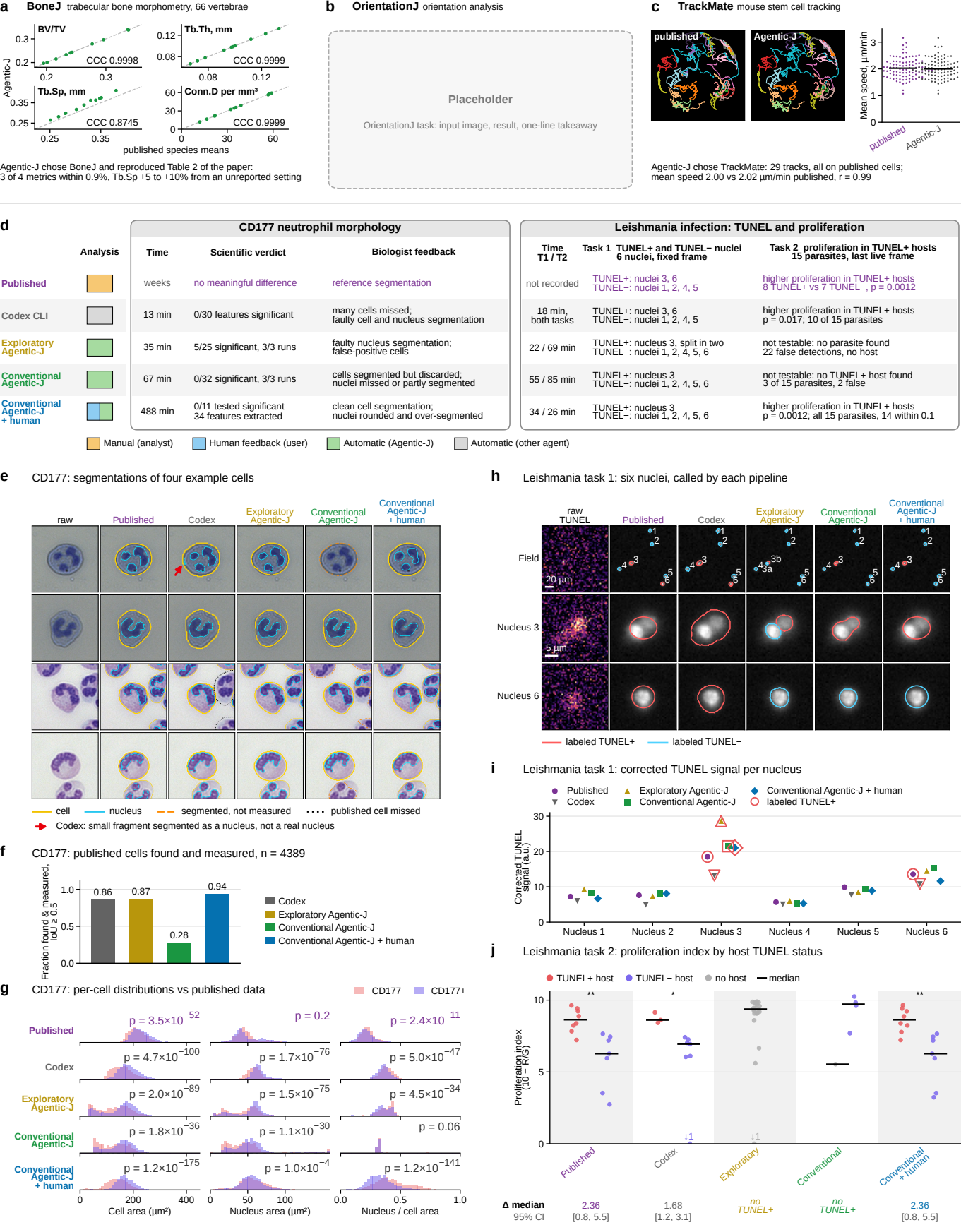

a
BoneJ trabecular bone morphometry, 66 vertebrae
BV/TV
Tb.Th, mm
Tb.Sp, mm
Conn.D per mm³
CCC 0.9998
CCC 0.9999
CCC 0.8745
CCC 0.9999
Agentic-J
published species means
Agentic-J chose BoneJ and reproduced Table 2 of the paper:
3 of 4 metrics within 0.9%, Tb.Sp +5 to +10% from an unreported setting
b
OrientationJ orientation analysis
Placeholder
OrientationJ task: input image, result, one-line takeaway
c
TrackMate mouse stem cell tracking
published
Agentic-J
Mean speed, µm/min
Agentic-J chose TrackMate: 29 tracks, all on published cells;
mean speed 2.00 vs 2.02 µm/min published, r = 0.99
d
Analysis
CD177 neutrophil morphology
Time
Scientific verdict
Biologist feedback
Published
weeks
no meaningful difference
reference segmentation
Codex CLI
13 min
0/30 features significant
many cells missed;
faulty cell and nucleus segmentation
Exploratory Agentic-J
35 min
5/25 significant, 3/3 runs
faulty nucleus segmentation;
false-positive cells
Conventional Agentic-J
67 min
0/32 significant, 3/3 runs
cells segmented but discarded;
nuclei missed or partly segmented
Conventional Agentic-J + human
488 min
0/11 tested significant
34 features extracted
clean cell segmentation;
nuclei rounded and over-segmented
Leishmania infection: TUNEL and proliferation
Time T1 / T2
Task 1 TUNEL+ and TUNEL− nuclei 6 nuclei, fixed frame
Task 2 proliferation in TUNEL+ hosts 15 parasites, last live frame
not recorded
TUNEL+: nuclei 3, 6
TUNEL−: nuclei 1, 2, 4, 5
higher proliferation in TUNEL+ hosts
8 TUNEL+ vs 7 TUNEL−, p = 0.0012
18 min, both tasks
TUNEL+: nuclei 3, 6
TUNEL−: nuclei 1, 2, 4, 5
higher proliferation in TUNEL+ hosts
p = 0.017; 10 of 15 parasites
22 / 69 min
TUNEL+: nucleus 3, split in two
TUNEL−: nuclei 1, 2, 4, 5, 6
not testable: no parasite found
22 false detections, no host
55 / 85 min
TUNEL+: nucleus 3
TUNEL−: nuclei 1, 2, 4, 5, 6
not testable: no TUNEL+ host found
3 of 15 parasites, 2 false
34 / 26 min
TUNEL+: nucleus 3
TUNEL−: nuclei 1, 2, 4, 5, 6
higher proliferation in TUNEL+ hosts
p = 0.0012; all 15 parasites, 14 within 0.1
Manual (analyst)
Human feedback (user)
Automatic (Agentic-J)
Automatic (other agent)
e
CD177: segmentations of four example cells
raw
Published
Codex
Exploratory Agentic-J
Conventional Agentic-J
Conventional Agentic-J + human
cell
nucleus
segmented, not measured
published cell missed
Codex: small fragment segmented as a nucleus, not a real nucleus
f
CD177: published cells found and measured, n = 4389
Fraction found & measured, IoU ≥ 0.5
0.86
0.87
0.28
0.94
Codex
Exploratory Agentic-J
Conventional Agentic-J
Conventional Agentic-J + human
g
CD177: per-cell distributions vs published data
CD177−
CD177+
Published
Codex
Exploratory Agentic-J
Conventional Agentic-J
Conventional Agentic-J + human
p = 3.5×10⁻⁵²
p = 0.2
p = 2.4×10⁻¹¹
p = 4.7×10⁻¹⁰⁰
p = 1.7×10⁻⁷⁶
p = 5.0×10⁻⁴⁷
p = 2.0×10⁻⁸⁹
p = 1.5×10⁻⁷⁵
p = 4.5×10⁻³⁴
p = 1.8×10⁻³⁶
p = 1.1×10⁻³⁰
p = 0.06
p = 1.2×10⁻¹⁷⁵
p = 1.0×10⁻⁴
p = 1.2×10⁻¹⁴¹
Cell area (µm²)
Nucleus area (µm²)
Nucleus / cell area
h
Leishmania task 1: six nuclei, called by each pipeline
raw TUNEL
Published
Codex
Exploratory Agentic-J
Conventional Agentic-J
Conventional Agentic-J + human
Field
Nucleus 3
Nucleus 6
20 µm
5 µm
labeled TUNEL+
labeled TUNEL−
i
Leishmania task 1: corrected TUNEL signal per nucleus
Published
Codex
Exploratory Agentic-J
Conventional Agentic-J
Conventional Agentic-J + human
labeled TUNEL+
Corrected TUNEL signal (a.u.)
Nucleus 1
Nucleus 2
Nucleus 3
Nucleus 4
Nucleus 5
Nucleus 6
j
Leishmania task 2: proliferation index by host TUNEL status
TUNEL+ host
TUNEL− host
no host
median
Proliferation index (10 − R/G)
Published
Codex
Exploratory
Conventional
Conventional + human
Δ median
95% CI
2.36
[0.8, 5.5]
1.68
[1.2, 3.1]
no TUNEL+
no TUNEL+
2.36
[0.8, 5.5]